INCH-POUND

MIL-DTL-16377/57B(SH)
2 August 1996
SUPERSEDING
MIL-F-16377/57A(SH)
9 May 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING, 51, 68 AND 102 WATTS, 120 VOLTS, 60 HERTZ, DRIPPROOF, SYMBOLS 348, 349 AND 350

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-16377 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

All dimensions in inches.
For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

AMSC N/A FSC 6210
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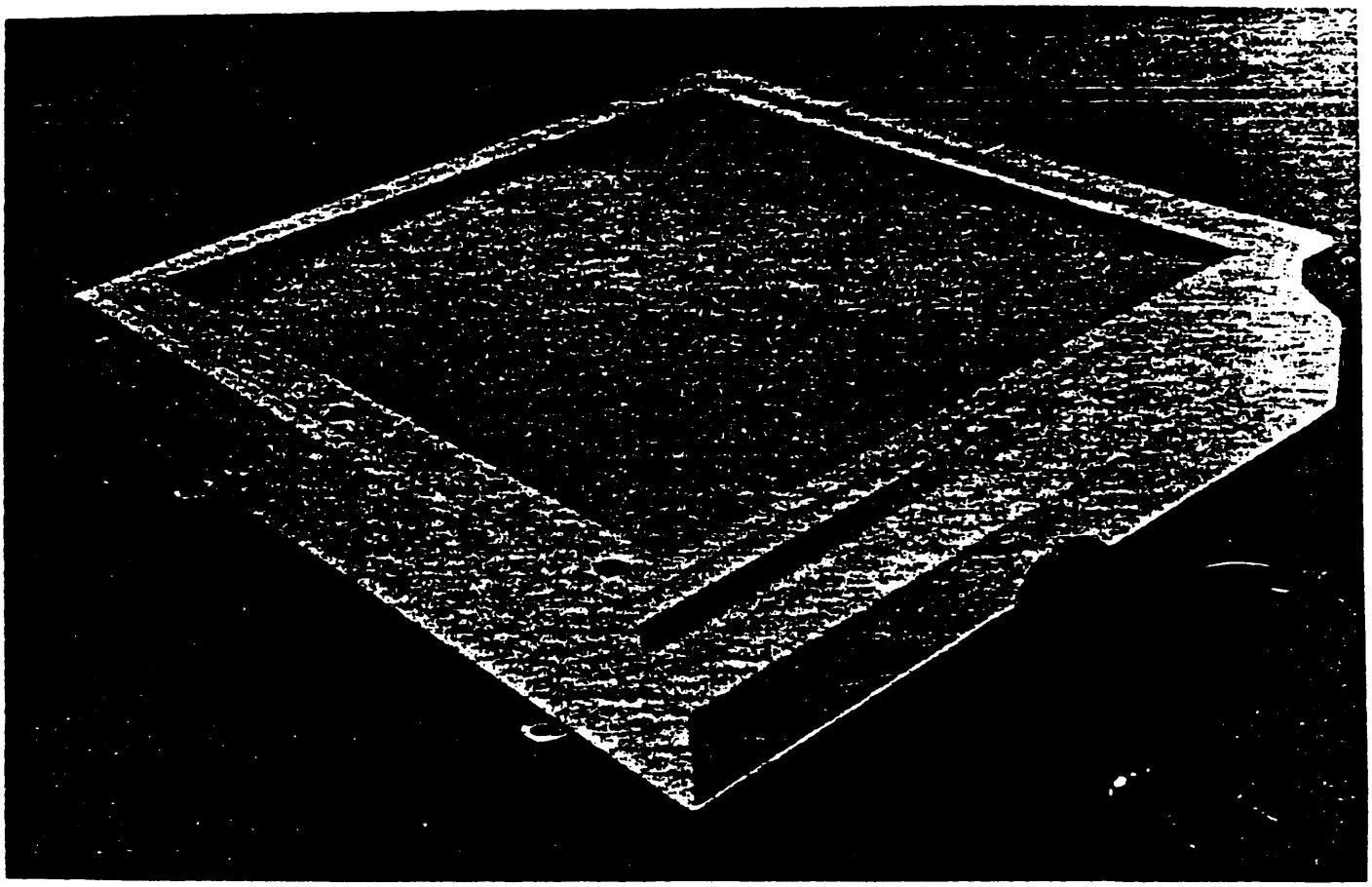
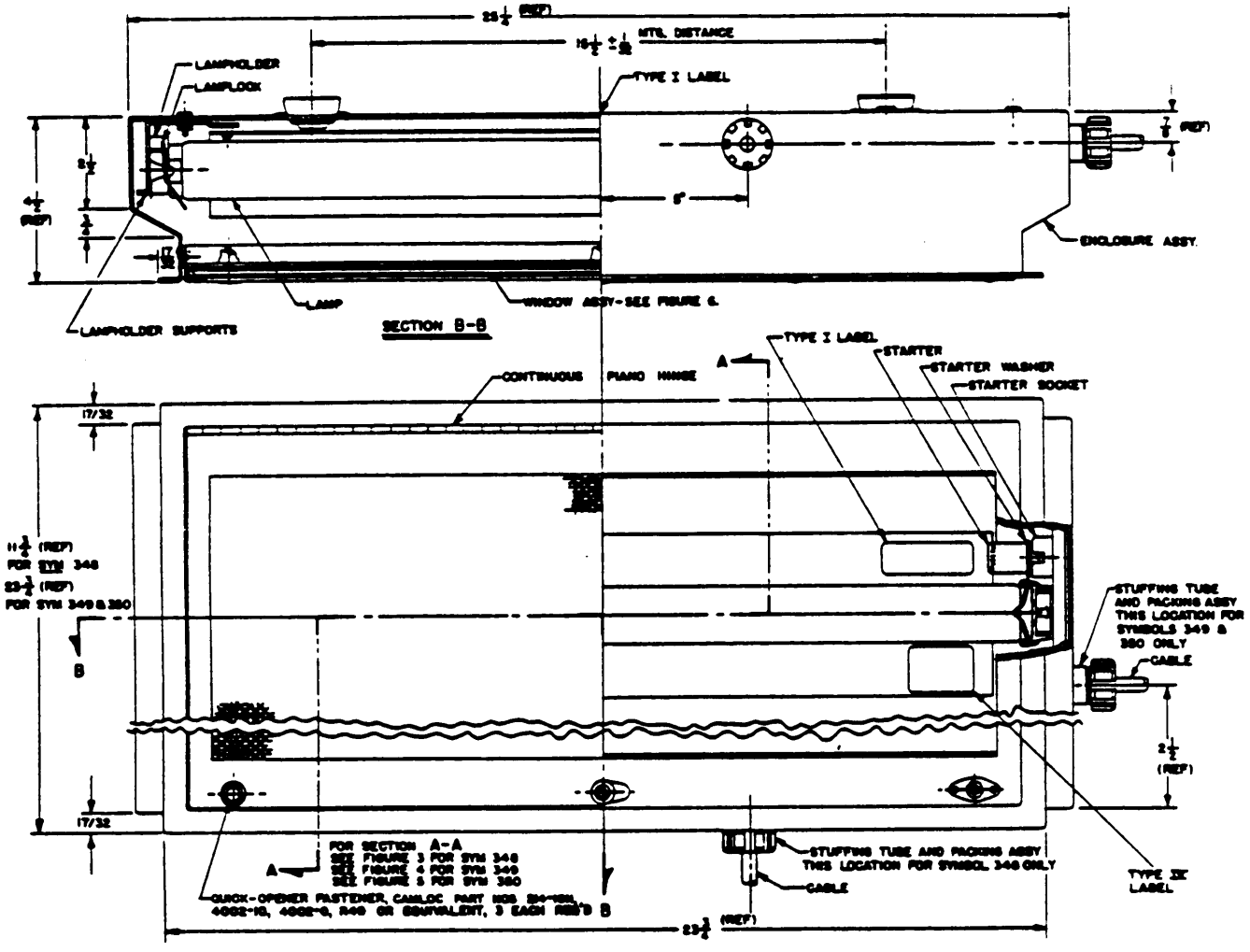


FIGURE 1. Typical view of Symbols 349 and 350.

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NOTES: Dimensions shown may be modified to accommodate ballast.
Critical dimensions (mounting, window area, lamp spacing, etc.) must be maintained.

FIGURE 2. Typical view of Symbols 348, 349 and 350.

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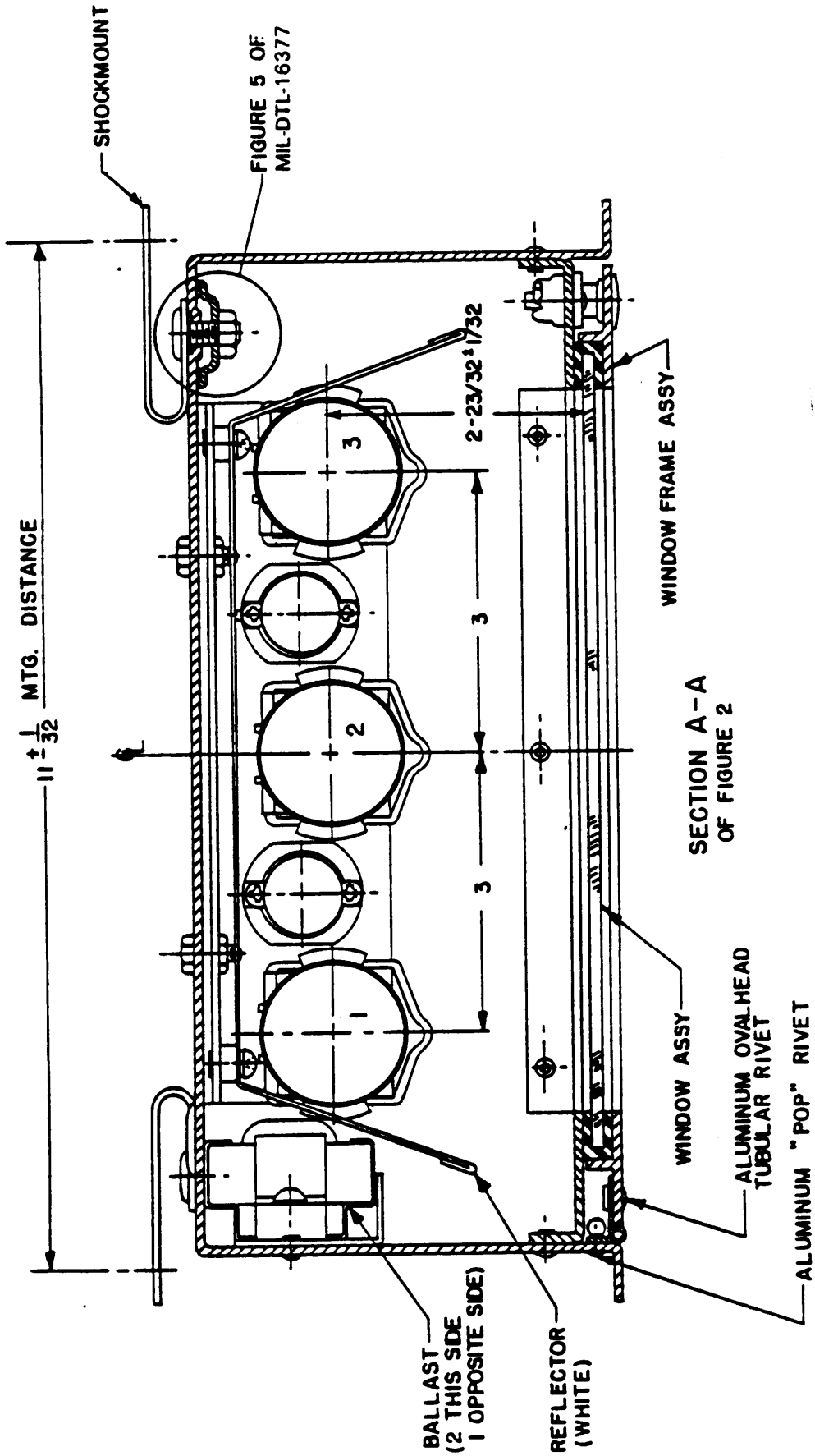


FIGURE 3. Typical section view of Symbol 348.

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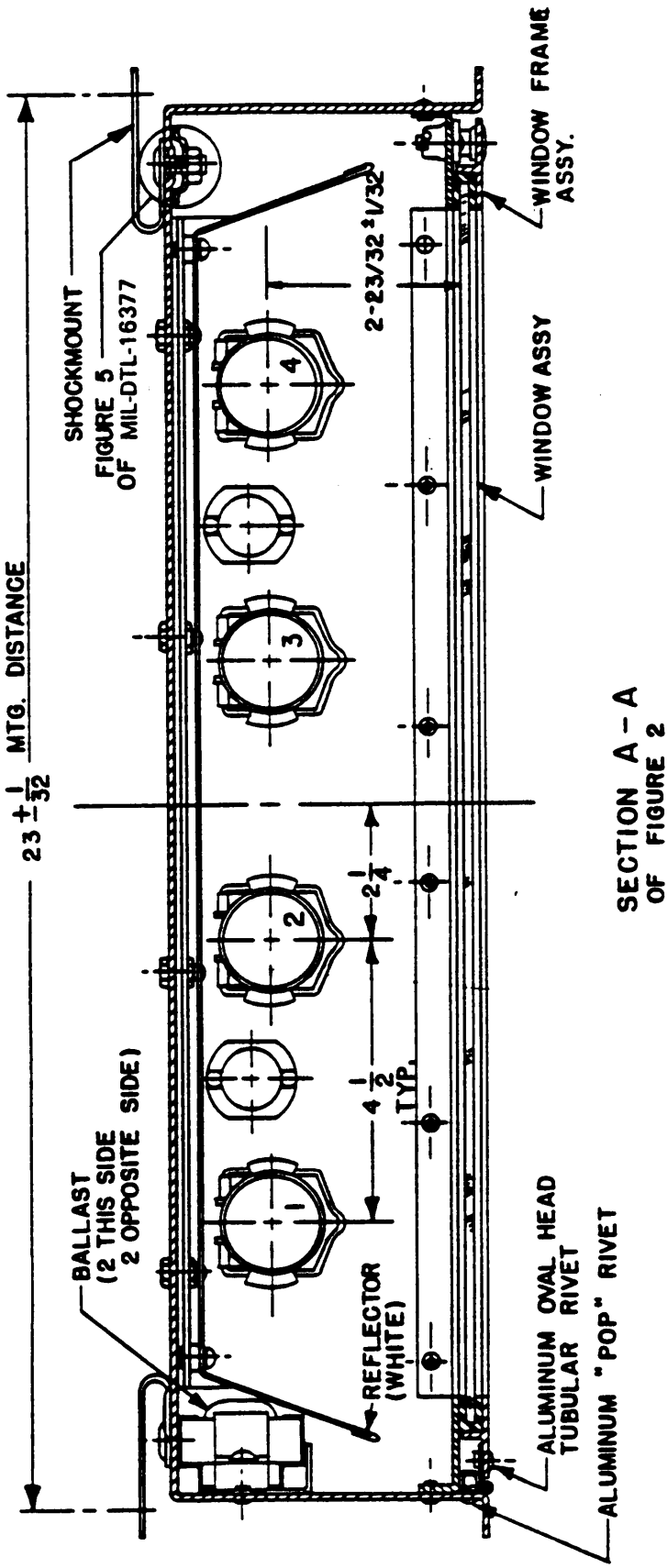
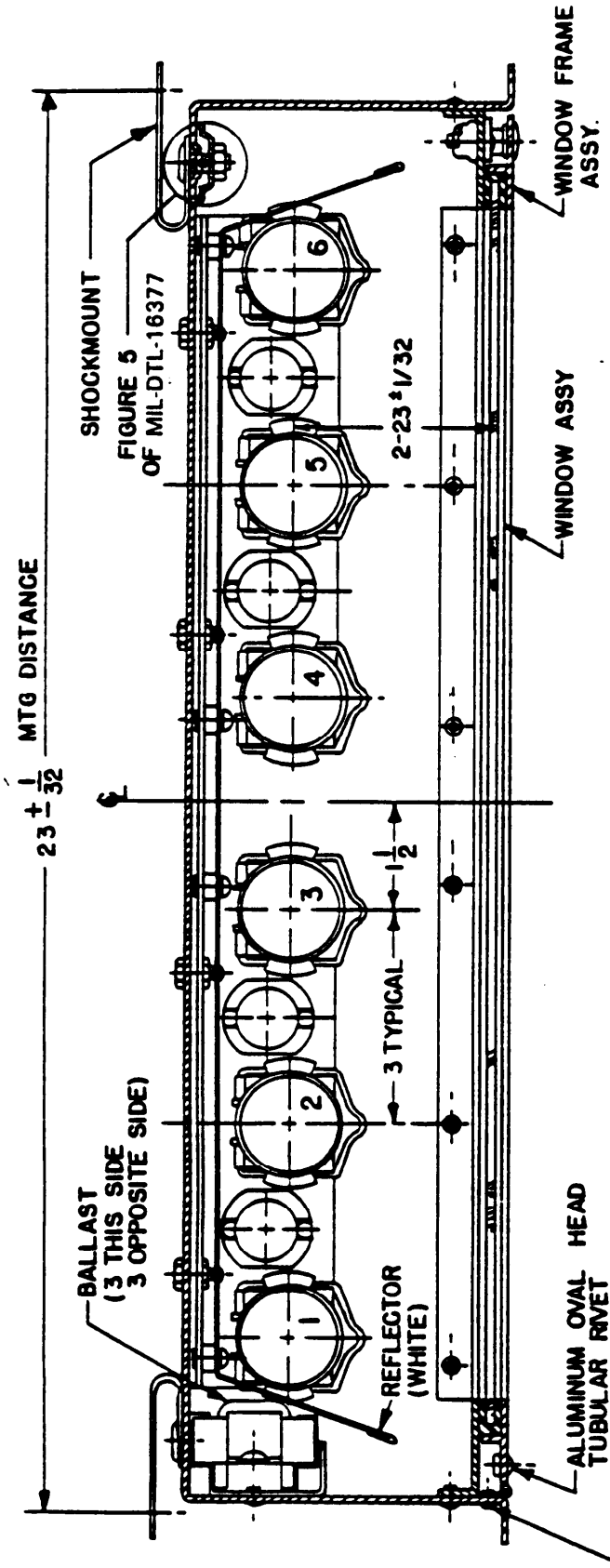


FIGURE 4. Typical section of Symbol 349.

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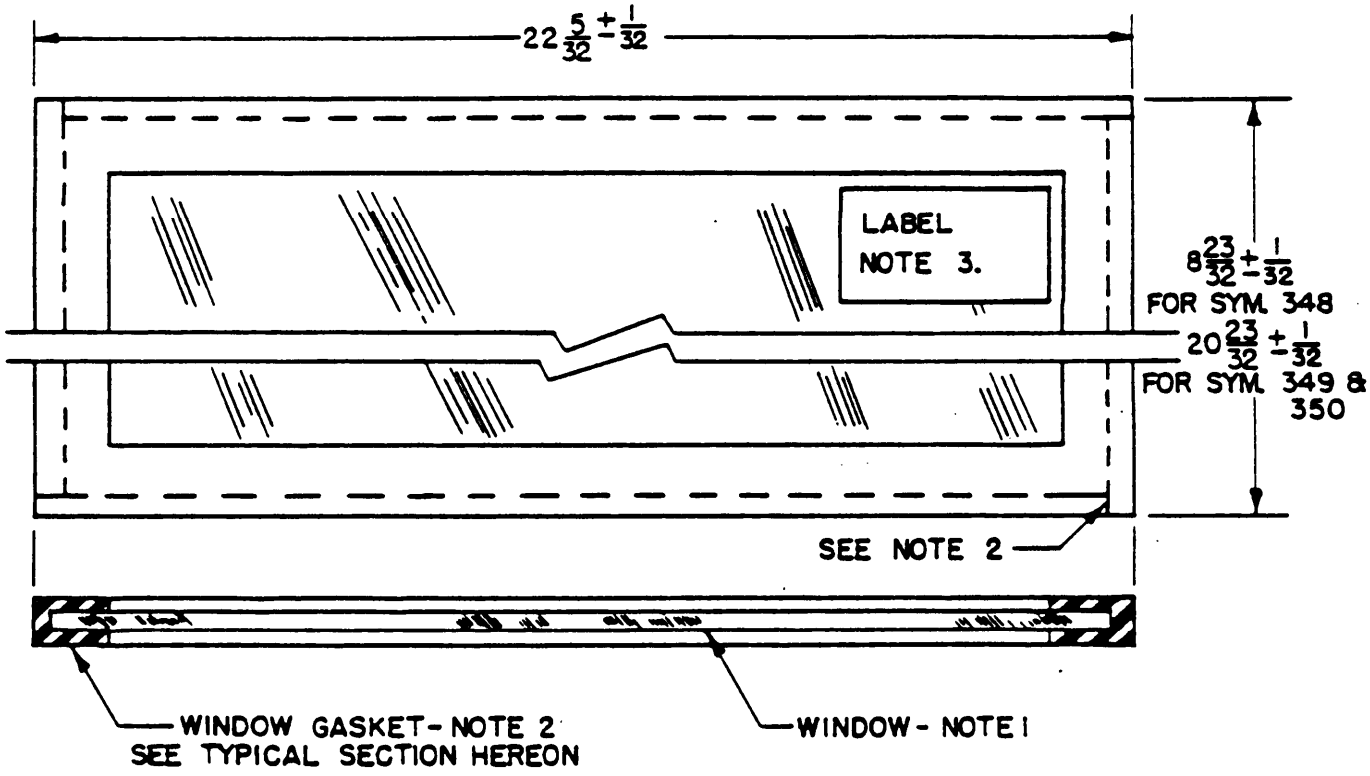


SECTION A-A
OF FIGURE 2

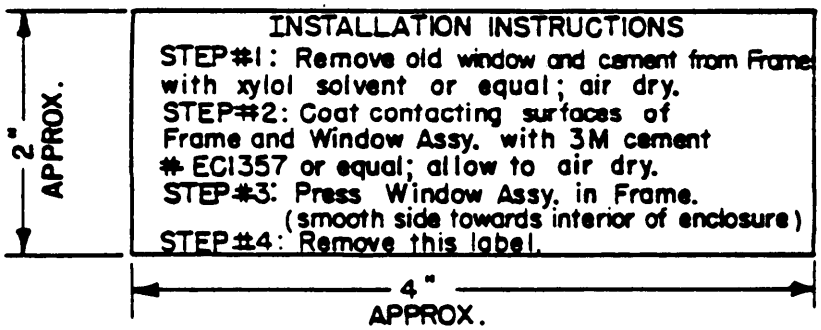
NOTE: For red and white illumination installing activity shall provide red filter tubes on #2 and #5 lamps.

FIGURE 5. Typical section of Symbol 350.

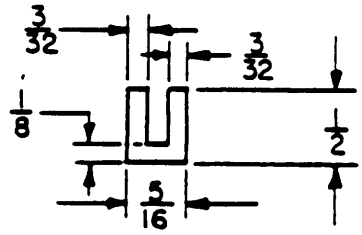
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(a) Window Assembly



(b) LABEL - Note 3.



STANDARD RMA TOLERANCES FOR EXTRUSION SHALL APPLY

(c) GASKET - Typical Section

NOTES:

1. Material for window shall be 1/8 inch thick, clear, prismatic (one side only), heat resistant, cast acrylic plastic sheet, type CLEAR-2 of MIL-P-24191.
2. Material for window gasket shall be neoprene rubber 25-35 durometer. Gasket shall be cemented to window. Gasket may be furnished in segments.
3. For window assemblies furnished as repair parts, a type IV label shall be furnished in accordance with MIL-DTL-16377. Location and the information to be shown on this label shall be as shown hereon.

FIGURE 6. Wind

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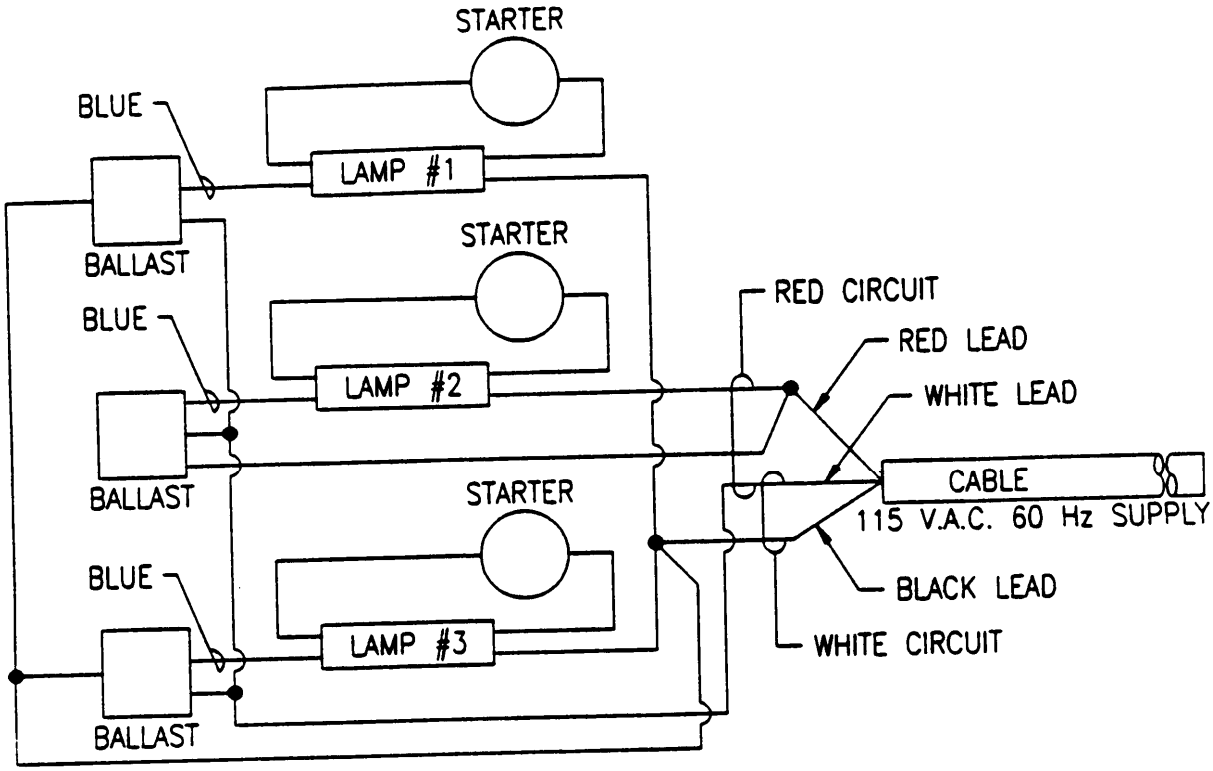


FIGURE 7. Wiring diagram for Symbol 348.

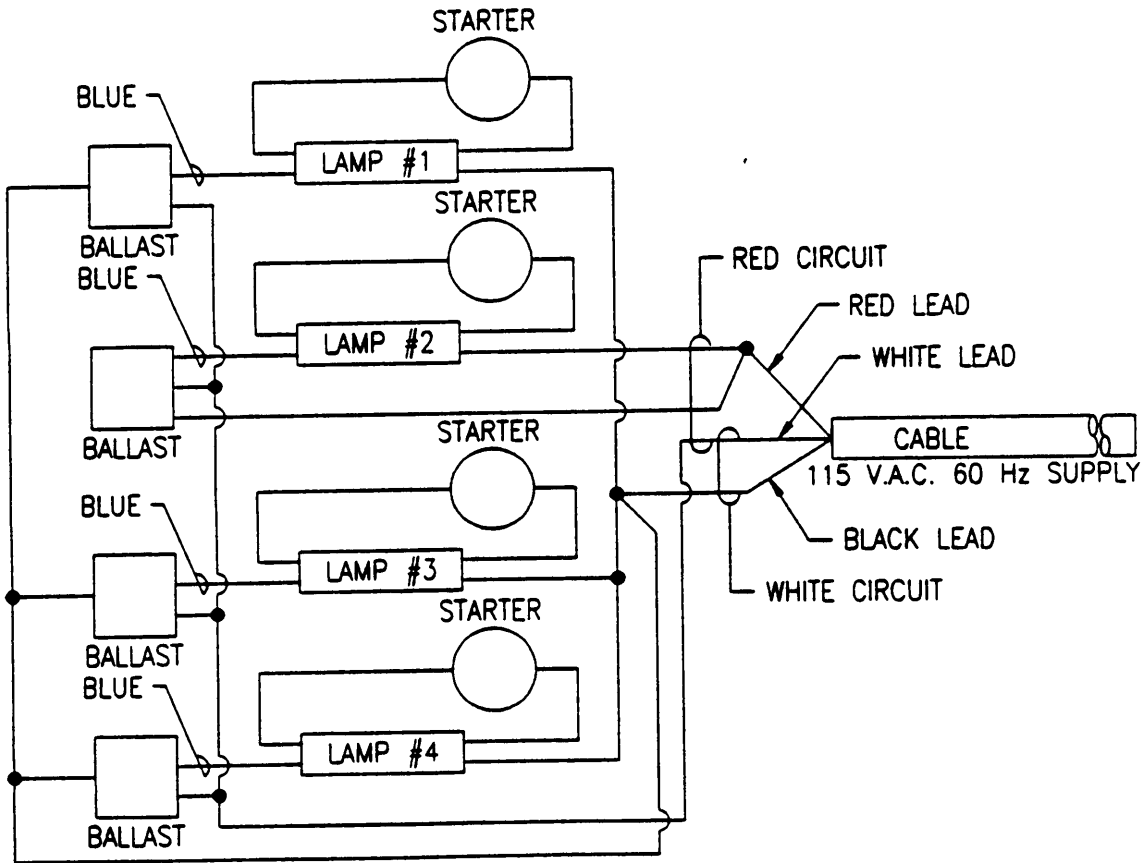


FIGURE 8. Wiring diagram for Symbol 349.

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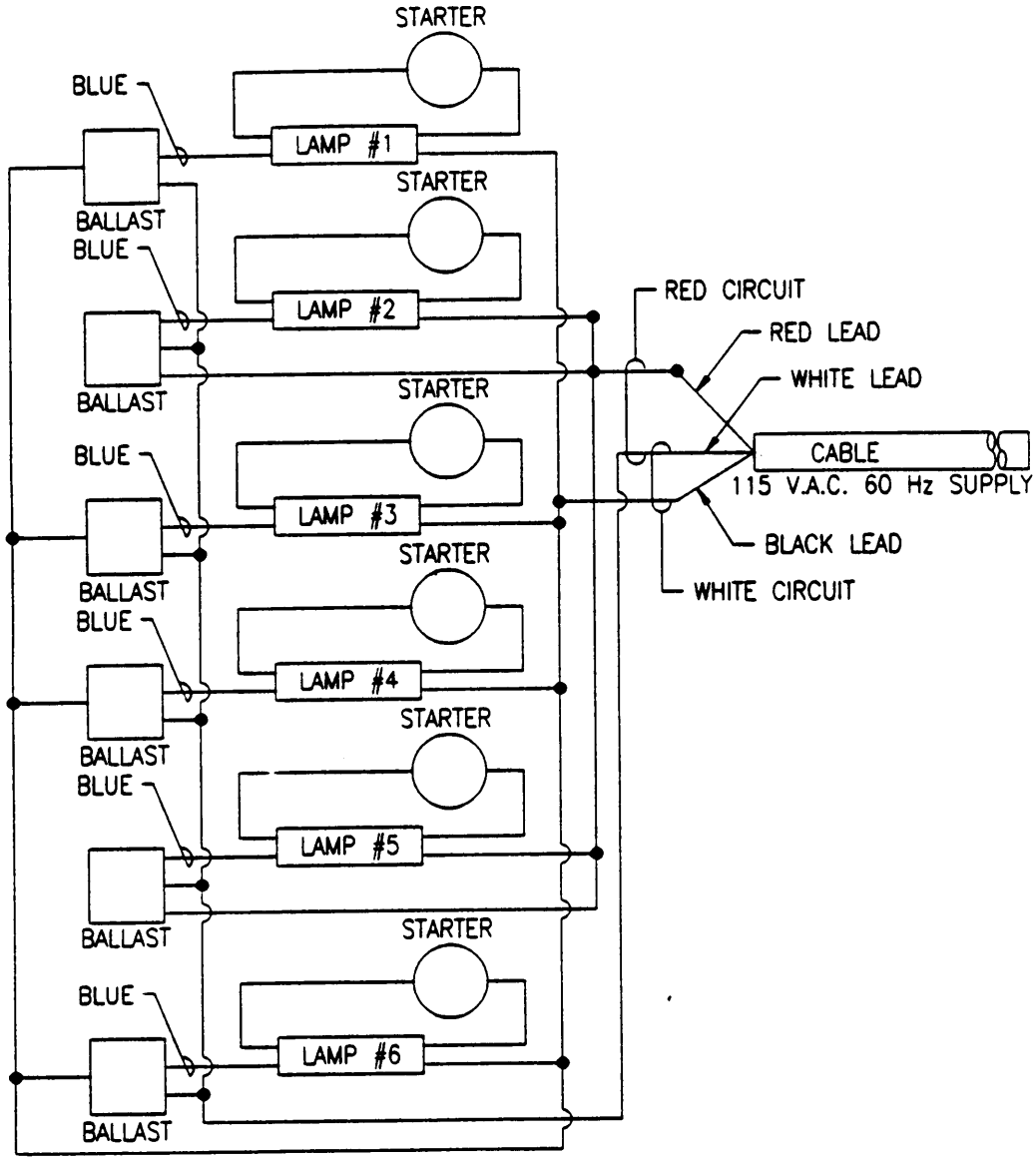
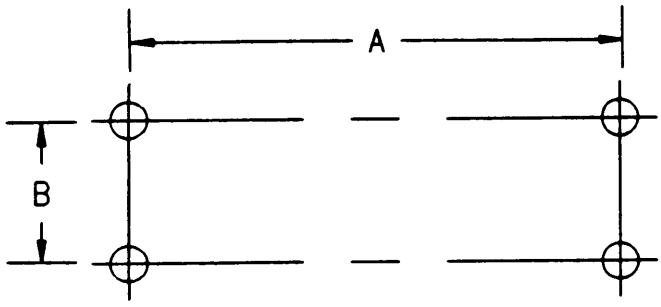


FIGURE 9. Wiring diagram for Symbol 350.

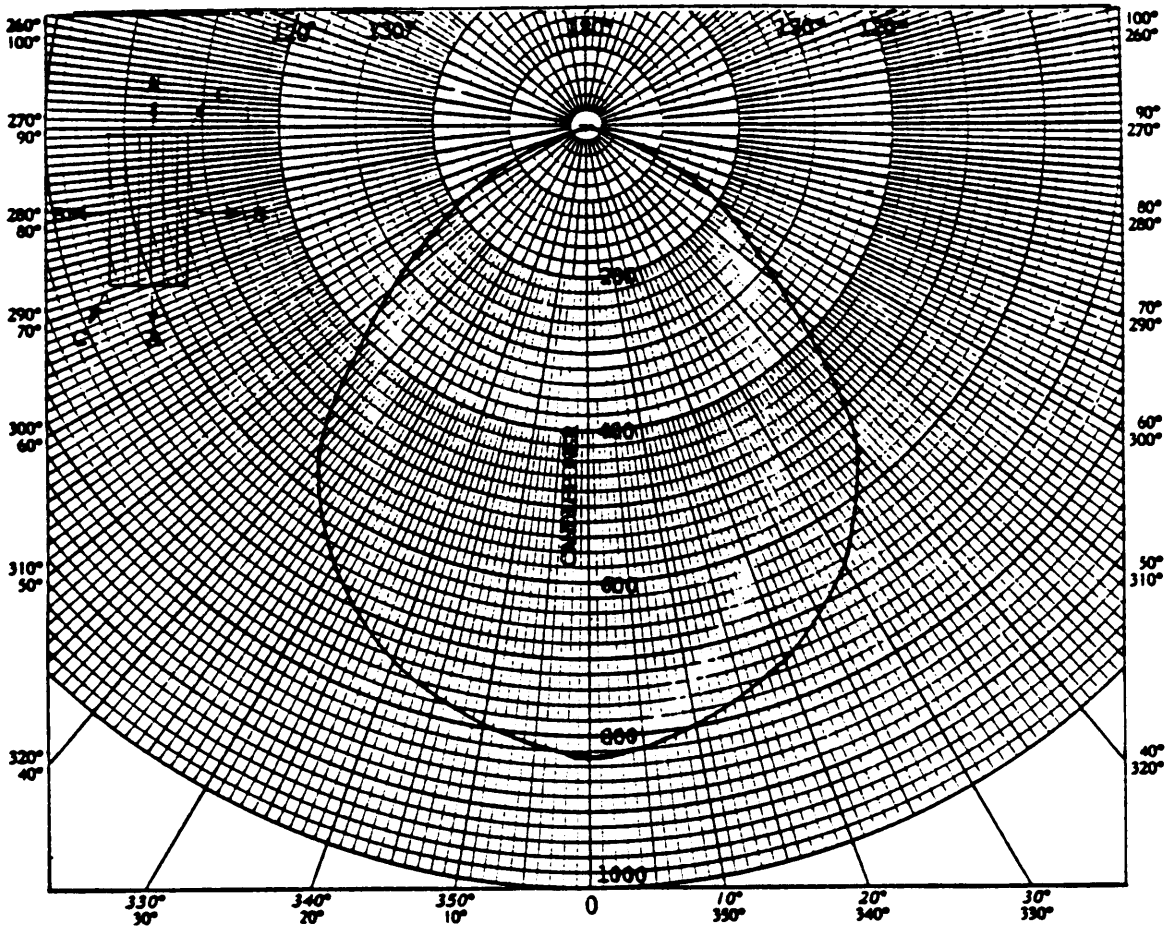


SYMBOL	DIMENSION	
	A	B
348	11	15.5
349, 350	23	15.5

MOUNTING BOLTS SHALL BE 3/8-16

FIGURE 10. Mounting dimersions.

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ANGLE	CANDLEPOWER* SYMBOL 348
0	825
5	810
10	800
15	760
20	750
25	715
30	670
35	620
40	550
45	440
50	330
55	240
60	180

ANGLE	CANDLEPOWER* SYMBOL 348
65	130
70	110
75	90
80	45
85	20
90	10
95	5
100	0
105	0
110	0
115	0
120	0
125	0

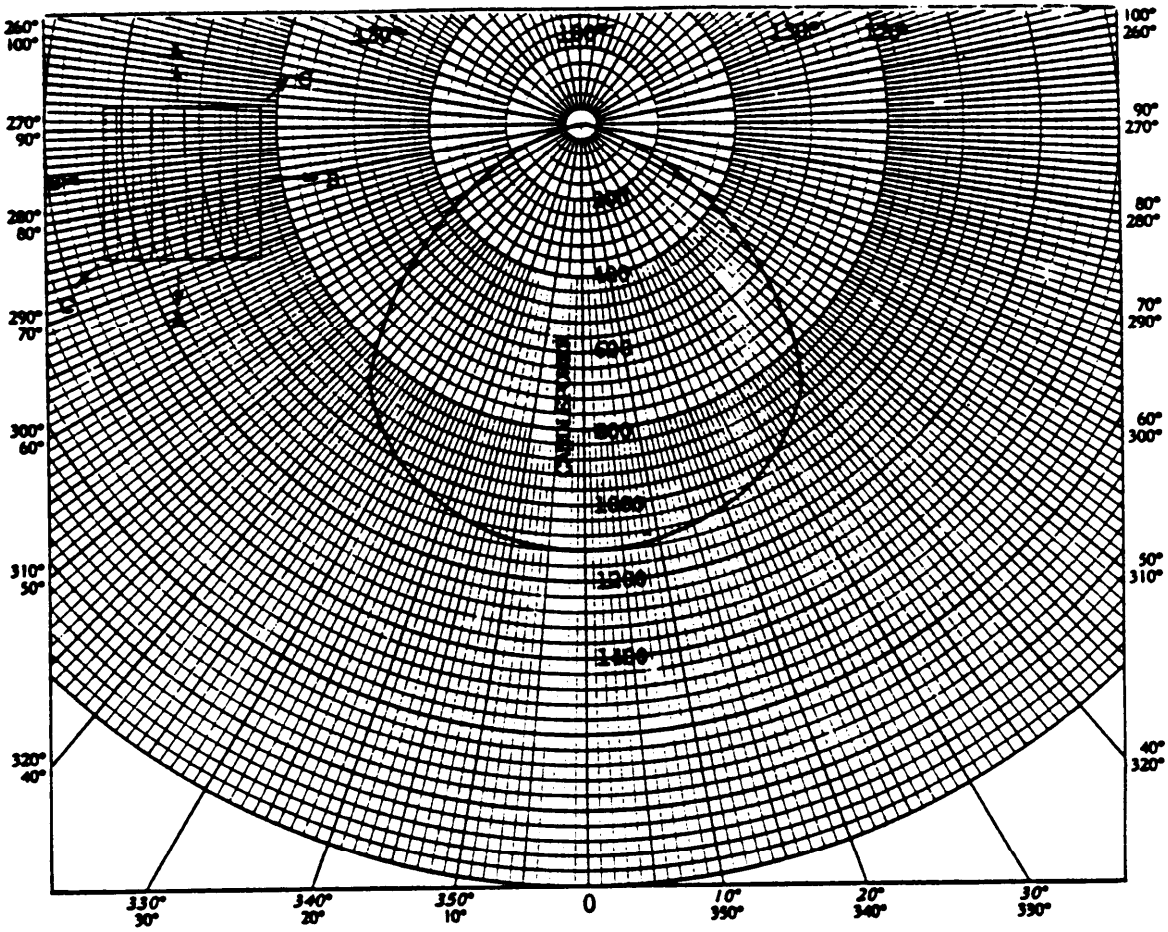
*Average values for planes A-A, B-B and C-C.

Photometric test data:

Distance - - - 10 feet.
Plane - - - - At right angles to long axis of fixture through geometric center of lamp.
Voltage - - - 115 volts, 60 Hz.
Ambient temperature - - - 76°F
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp Correction: - - - Distribution curves based on 1280 lumens per lamp.

FIGURE 11. Candlepower distribution cuvves for Symbol 348.

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ANGLE	CANDLEPOWER* SYMBOL 349
0	1120
5	1120
10	1100
15	1080
20	1070
25	1035
30	1000
35	935
40	890
45	825
50	600
55	450
60	275

ANGLE	CANDLEPOWER* SYMBOL 349
65	220
70	165
75	55
80	45
85	25
90	0
95	0
100	0
105	0
110	0
115	0
120	0
125	0

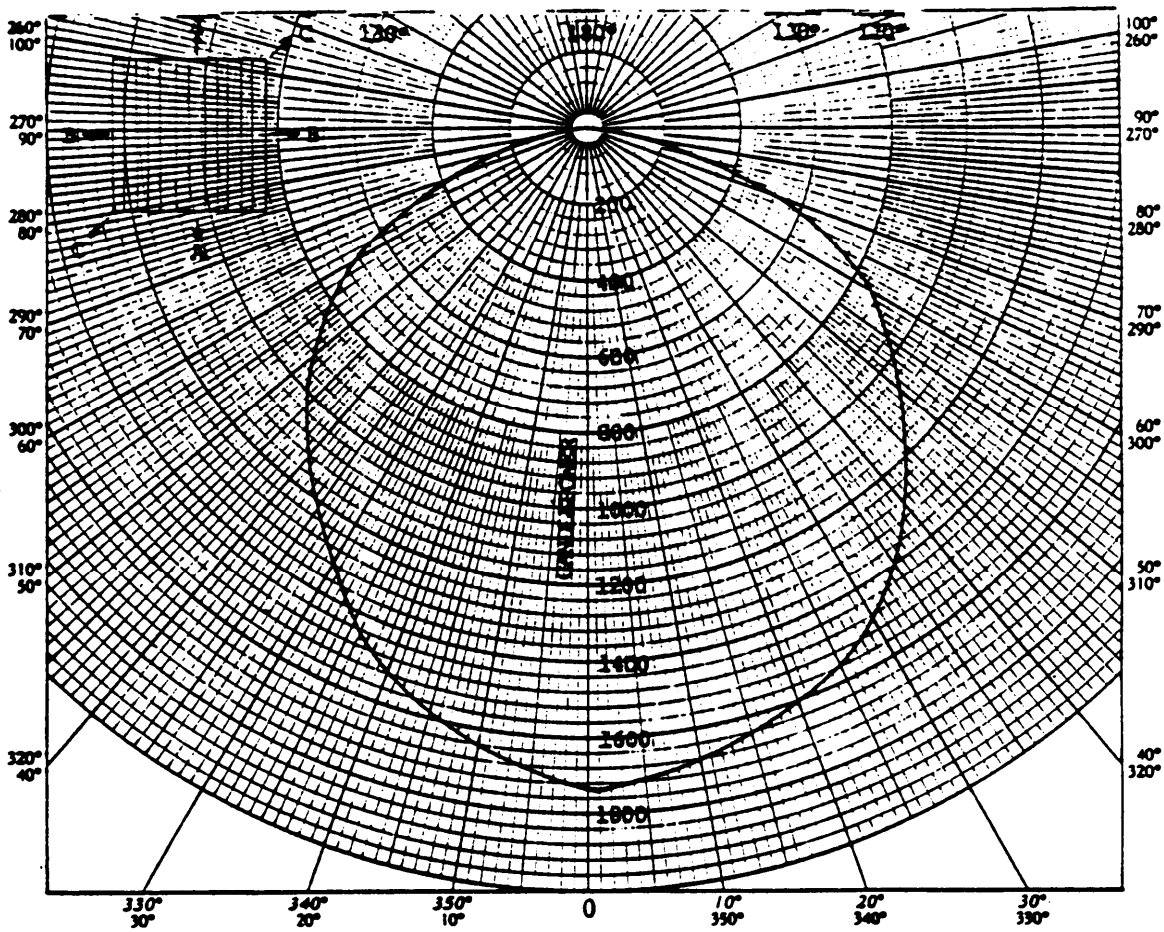
*Average values for planes A-A, B-B and C-C.

Photometric test data:

Distance - - - 10 feet.
Plane - - - - At right angles to long axis of fixture through geometric center of lam
Voltage - - - 115 volts, 60 Hz.
Ambient temperature - - - 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp Correction: - - - Distribution curves based on 1280 lumens per lamp.

FIGURE 12. Candlepower distribution curves for Symbol 349.

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ANGLE	CANDLEPOWER* SYMBOL 350
0	1760
5	1760
10	1740
15	1720
20	1670
25	1630
30	1550
35	1475
40	1365
45	1255
50	880
55	660
60	440

ANGLE	CANDLEPOWER* SYMBOL 350
65	330
70	220
75	110
80	55
85	25
90	0
95	0
100	0
105	0
110	0
115	0
120	0
125	0

*Average values for planes A-A, B-B and C-C.

Photometric test data:

Distance - - - 10 feet.
Plane - - - - At right angles to long axis of fixture through geometric center of lamp.
Voltage - - - 115 volts, 60 Hz.
Ambient temperature - - - 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp Correction: - - - Distribution curves based on 1280 lumens per lamp.

FIGURE 13. Candlepower distribution curves for symbol 350.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
348: Fixture, flush mounting, with three 17-watt cool white fluorescent lamps and clear prismatic window.
349: Fixture, flush mounting, with four 17-watt cool white fluorescent lamps and clear prismatic window.
350: Fixture, flush mounting, with six 17-watt cool white fluorescent lamps and clear prismatic window.

DESIGN:
Electrical rating: 51, 68, and 102 watts, 120 volts, 60 Hertz.
Dimension and configurations: See figures 1 through 5.
Distribution curves: Candlepower - See figures 11, 12, and 13.
Cable: 10 feet of type LSTHOF-3 in accordance with MIL-C-24643/3.
Wiring diagram: See figures 7, 8, and 9.
Window assembly: See figure 6.
Mounting dimensions: See figure 10.
Enclosure: Dripproof 15 degrees.
Weight:
Symbol 348: 20 lbs maximum.
Symbol 349: 26 lbs maximum.
Symbol 350: 30 lbs maximum.

Designation and marking: Labels shall be furnished in accordance with the following types of MIL-DTL-16377:
(a) Type I.
(b) Type IV - One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I by name and stock number only.

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:
Complete fixtures: See table II.
Window assemblies: See table III.

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Conformance:
Comparison inspection: Same as the first article inspection.
Inspection of product for delivery:
Examination: See MIL-DTL-16377.
Examination for preparation of delivery: See MIL-DTL-16377.
Conformance tests: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

TABLE I. Standard parts.

Name	Specification or Military part number	National stock number (NSN)	Remarks
Lamp	A-A-50597		Industry number F17T12/CW
Lamp lock	M16377/40-002	6250-00-399-7040	
Lampholder	UL-542	6250-00-284-0481	SUPP. SA, FIG. SA2.2
Startersocket	UL-542	6250-00-295-2738	SUPP. SA, FIG. SA2.4
Starter	20W, type III of W-S-755	6250-00-299-2884	Commercial designation: FS2
Ballast	M16377/72-001		
Shockmount: Symbol 348 Symbols 349, 350	M16377/41-009 or M16377/41-008	6210-00-973-1747	See 3.5.16 of MIL-DTL-16377
Red filter tube	M16377/42-003	6210-00-295-2838	Not furnished. See 3.7.7 of MIL-DTL-16377

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TABLE II. First article and comparison inspection for complete fixtures.

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.2	Dripproof 15 degrees
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Dripproof 15 degrees
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Electromagnetic interference	3.6.14	4.8.16	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Light output	3.7.1	4.8.6	Candlepower
Magnetic permeability	3.4.8	4.8.18	

TABLE III. First article and comparison inspection for window assemblies.

Test	Requirement	Test method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	

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TABLE IV. Part number and general information.

Military part number M16377/57-	Item name	National stock number (NSN)	Remarks
348	Complete fixture, Symbol 348	Not assigned	
349	Complete fixture, Symbol 349	Not assigned	
350	Complete fixture, Symbol 350	Not assigned	
001	Window assembly for Symbol 348		Figure 6
002	Window assembly for Symbols 349 and 350		Figure 6

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6210-N661-09)

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